

Bhumika Gupta

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Available for co-op terms starting Jan 2027 (4-8 months)

EDUCATION

McMaster University

Hamilton, ON

Bachelor of Applied Science in Computer Science

Sep. 2023 – Apr. 2028

Relevant Coursework: Data Structures and Algorithms, Algorithms and Complexity, Software Development, Databases, Operating Systems, Networks and Security, Software Testing

EXPERIENCE

Application/Software Developer Co-op

May 2026 – Dec. 2026

CIBC

Toronto, ON

- Develop Python automation and data-processing tools supporting daily Securities Lending trading and operations
- Migrate 15+ Excel/VBA workflows to scheduled Python services, eliminating recurring manual task execution
- Implement validation and exception handling logic to improve accuracy and reliability for trader reporting
- Debug trade-reporting data discrepancies in SQL Server, validating data and refining queries for accurate reporting

Test Analyst Co-op

Sep. 2025 – Dec. 2025

CIBC

Toronto, ON

- Integrated a synthetic data generation utility across Databricks, Oracle, Netezza, and Azure environments
- Validated data across cloud and on-prem databases using SQL, Databricks, Toad, and Excel VBA
- Designed test cases from technical requirements and collaborated with developers to debug data pipelines

AI Researcher & Program Strategist

June 2025 – Aug. 2025

ELM Tree

Toronto, ON

- Researched AI adoption and workforce trends to inform internal product and program strategy

Web Developer

June 2024 – Aug. 2024

ELM Tree

Toronto, ON

- Redesigned and implemented responsive web interfaces in Figma and Wix, increasing user engagement by 15%

PROJECTS

Quoted: Book Quote Web Scraper | *Python, Flask, React, httpx, selectolax*

- Built a full-stack application that scrapes, stores, and displays book excerpts from public websites
- Developed Flask API endpoints and a scraping pipeline with HTML parsing, deduplication, and paginated results

Pathfinding Algorithm Suite | *Python, Pandas, Matplotlib*

- Implemented and benchmarked Dijkstra's, Bellman-Ford, and A* on London Underground geolocation data
- Designed a modular OOP architecture with custom heuristics and performance visualization

Geometric Shape Testing Framework | *Java, JUnit*

- Developed JUnit tests for geometric collision detection, edge cases, and invalid-input handling
- Applied modular OOP design to build reusable geometry and validation logic

Blast From The Past | *Elm*

- Built an interactive Elm game prototype exploring Parkinson's disease assessment, using iterative user feedback

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C, Scala, Elm, HTML/CSS

Libraries/Frameworks: React, Next.js, Node.js, Flask, Pandas, Matplotlib

Tools: Git, Databricks, SQL Server/SSMS, Toad for Oracle, Azure Storage, Db2, DBeaver, Excel VBA, Figma